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Operations and Supply Chain Management

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Uncharted Territory: Exploring the Unknowns of AI in Operations and Supply Chains

I have been lucky this semester. Most students writing about AI in operations have to work from articles and case studies; I have spent the last four months on the inside of a company wrestling with these questions in real time. It has given me some great insight into how a real company is adapting to this new and developing AI revolution. As a Digital and AI Strategy Intern at one of the world's largest privately owned insurance brokerages, my job was to interview people across the business in areas like HR, broking, IT, change management, and governance about how AI is actually changing what they do. This course mostly focused on physical operations, inventories, cycle times, line balancing, MRP, etc. and that angle remains relevant. But what I have come to believe is that the most uncharted territory in operations today is in service operations, the chapter that turns out to be most directly relevant to where AI is actually landing. Services have characteristics that distinguish them from manufacturing: they cannot be stored in inventory, capacity becomes the dominant issue, and the customer is involved in the process itself. AI is the first technology in decades to seriously challenge all three of those assumptions at once, and an insurance brokerage is one of the clearest places to see it happen.

Starting with the service package framework, an insurance brokerage is overwhelmingly an information business and a trust business. The supporting facility and facilitating goods are

minimal. What the client is buying is information, the broker's knowledge of the market, and the implicit services of relationship and judgement. As regards broking at Lockton, this activity occupies a prominent position in the high-end of customer contact and face-to-face interaction, total customization of the process, and hence, a point where sales possibilities are maximum and efficiency levels minimum in terms of the design matrix of service systems. Currently, AI technologies are making this compromise very difficult. The technologies being tried at the company try to shift some processes towards the other side of the design matrix – the buffered core – where efficiency levels are higher but without any sacrifice in relationship advantages owing to high customer interaction.

The course tool I have found most useful for thinking about what is happening is the service blueprint, with its line of visibility separating what the customer sees from what they do not. A broker's process has clear back-stage activities: pulling submission data together, drafting renewal reports, and building what one of my coworkers in the Marine team called a "mud map," the layered tower of insurers that makes up a complex placement. AI is being aimed almost entirely at this back-stage region. The firm's main internal AI program that is being developed, called Project Polaris, projects a forty to fifty percent time saving in the Risk Intake workflow alone, achieved by automating data entry and using agentic AI for higher-value steps. In service-blueprint terms, this is a redesign of the back-stage process while leaving the front-stage interaction visibly unchanged. The same coworker who introduced project Polaris to me made a separate point that connects directly to fail safeing. He told me mistakes simply cannot be made in his line of work, and what he wanted from AI was a task tracker that would catch missed steps on complex policies. That is a poka-yoke in the precise sense the chapter defines it. The

unknown is whether AI poka-yokes, which are probabilistic rather than physical, can ever be as reliable as the limit switches and signal lamps the chapter describes in factory settings.

A second course concept that maps cleanly onto what I have seen is capacity. Because services cannot be stored in inventory, capacity is the dominant issue, and waiting time is one of the central problems service operations exist to manage. In broking, the equivalent of waiting time is the days or weeks between a client's submission and the issuance of a bound policy. A senior broker I spoke with who had been in the insurance industry for decades described an emerging digital facility called auto-follow, in which once a lead insurer has priced a risk, follow-market insurers can automatically take a share without each being individually pitched. Other tools aim to compress submission-to-bind from weeks to seconds through automated quote-and-bind systems. In service-operations terms, these are capacity expansions achieved not by adding servers but by changing the structure of the queue itself, removing entire phases of a multiphase service line. The unknown is whether the relationship-driven service that defines the firm survives the compression. The same broker raised the worry plainly: brokers, he said, will need to be mindful of being cut out of the supply chain entirely if pricing can be generated without them.

The third unknown is the one I came away from this semester most convinced about. New tools, on their own, do not change productivity. When personal computers arrived in the 1980s, economists predicted an immediate productivity boom. It did not come, at first, because firms were treating computers as expensive typewriters rather than redesigning their processes around them. The same dynamic is playing out now. There seemed to be a definite gap between the high-level vision being expressed about integration of AI processes, customized large language models, and agentic systems, and the reality at street level, where meetings were still

manually transcribed, holiday calculations were still done in Excel spreadsheets, and training completed in one system was not reflected in another system. As per a 2025 report from a major company in the insurance industry, less than seven percent of insurers had scaled up their use of AI, and the biggest reason cited for this failure was fragmentation of data and processes. The unknown here is not whether AI works; it is whether organizations have the operational discipline to do the unglamorous work of process redesign and data standardization that the technology requires.

AI in operations raises many uncertainties that are mainly found in the realm of services, areas for which our course's service operations module has given us terminology: the service package and what impact AI will have on its implicit elements, such as the service blueprint and line of visibility, capacity and queue management systems being undermined by automation, and fail-safing mechanisms requiring re-engineering in an era of probabilistic poka-yokes. The physical supply chain we covered will continue to play a role, but alongside this, there is a whole new range of operational challenges arising in the service sector with AI transforming the production process. Having spent four months inside a firm trying to work them out, I believe these are the right questions to be asking, and that the operations managers who will succeed are the ones willing to step into this uncharted territory rather than retreat from it.